

# SGPA and CGPA Calculation in C

## Algorithm Test Cases Program and Execution

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### 1 Aim and grading scale

To calculate each semester's credit-weighted SGPA and the cumulative credit-weighted CGPA for a variable number of semesters and courses, while validating every input.

| Grade       | O  | A | B | C | D | E | P |
|-------------|----|---|---|---|---|---|---|
| Grade point | 10 | 9 | 8 | 7 | 6 | 5 | 4 |

### 2 Algorithm

| Steps          | Procedure                                                                                                                                                                           |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 to 7         | Start. Declare the specified integer, character, and floating-point variables. Read and repeatedly validate positive global_id and m. Set cumulative_TC and cumulative_TGC to zero. |
| 8.1 to 8.3     | For each semester, reset semester_TC and semester_TGC. Read n and repeat until n is positive.                                                                                       |
| 8.4.1 to 8.4.6 | For each course, read and validate positive code and credit. Read grade and repeat until it is one of O, A, B, C, D, E, or P.                                                       |
| 8.4.7 to 8.4.9 | Convert grade to GP using the scale. Add GP * credit to semester_TGC and add credit to semester_TC.                                                                                 |
| 8.5 to 8.8     | Calculate SGPA = (float) semester_TGC / semester_TC. Display semester and SGPA. Add semester totals to cumulative totals.                                                           |
| 9 to 11        | Calculate CGPA = (float) cumulative_TGC / cumulative_TC. Display global_id, m, cumulative_TC, and CGPA. Stop.                                                                       |

### 3 Test case table

| Category            | Input                        | Expected                   | Observed | Status |
|---------------------|------------------------------|----------------------------|----------|--------|
| Valid two semesters | 2 semesters; 2 and 3 courses | SGPA 9.43, 8.22; CGPA 8.75 | Matched  | Pass   |
| Invalid global ID   | 0 then 1                     | Reject 0; accept 1         | Matched  | Pass   |
| Invalid m           | 0 then 1                     | Reject 0; accept 1         | Matched  | Pass   |
| Invalid n           | 0 then 1                     | Reject 0; accept 1         | Matched  | Pass   |
| Invalid code        | 0 then 101                   | Reject 0; accept 101       | Matched  | Pass   |
| Invalid credit      | 0 then 3                     | Reject 0; accept 3         | Matched  | Pass   |
| Invalid grade       | X then O                     | Reject X; accept O         | Matched  | Pass   |
| Boundary one course | m=1, n=1, credit=1, P        | SGPA and CGPA 4.00         | Matched  | Pass   |
| Highest grade       | one course, credit 3, O      | SGPA and CGPA 10.00        | Matched  | Pass   |
| Variable courses    | semester n values 1 and 3    | Correct weighted CGPA      | Matched  | Pass   |

### 4 C program

```
#include <stdio.h>
```

```
int main(void)
```

```
{
```

```
    int global_id, m, n, semester, course, code, credit, GP = 0;
```

SGPA and CGPA Calculation in C

```

int semester_TC, semester_TGC, cumulative_TC = 0, cumulative_TGC = 0;
char grade;
float SGPA, CGPA;

do {
    printf("Enter global ID: ");
    scanf("%d", &global_id);
    if (global_id <= 0)
        printf("Invalid global ID. Enter a positive value.\n");
} while (global_id <= 0);

do {
    printf("Enter number of semesters: ");
    scanf("%d", &m);
    if (m <= 0)
        printf("Invalid semester count. Enter a positive value.\n");
} while (m <= 0);

for (semester = 1; semester <= m; semester++) {
    semester_TC = 0;
    semester_TGC = 0;

    do {
        printf("\nEnter number of courses in semester %d: ", semester);
        scanf("%d", &n);
        if (n <= 0)
            printf("Invalid course count. Enter a positive value.\n");
    } while (n <= 0);

    for (course = 1; course <= n; course++) {
        do {
            printf("Enter code for course %d: ", course);
            scanf("%d", &code);
            if (code <= 0)
                printf("Invalid course code. Enter a positive value.\n");
        } while (code <= 0);

        do {
            printf("Enter credits for course %d: ", course);
            scanf("%d", &credit);
            if (credit <= 0)
                printf("Invalid credit. Enter a positive value.\n");
        } while (credit <= 0);

        do {
            printf("Enter grade for course %d (0/A/B/C/D/E/P): ", course);
            scanf(" %c", &grade);

            switch (grade) {
                case '0': GP = 10; break;
                case 'A': GP = 9; break;
                case 'B': GP = 8; break;
                case 'C': GP = 7; break;
                case 'D': GP = 6; break;
                case 'E': GP = 5; break;
                case 'P': GP = 4; break;
                default:
                    GP = 0;
                    printf("Invalid grade. Enter 0, A, B, C, D, E, or P.\n");
            }
        } while (GP == 0);

        semester_TGC += GP * credit;
        semester_TC += credit;
    }

    SGPA = (float) semester_TGC / semester_TC;
    printf("Semester %d SGPA = %.2f\n", semester, SGPA);

    cumulative_TGC += semester_TGC;
    cumulative_TC += semester_TC;
}

CGPA = (float) cumulative_TGC / cumulative_TC;
printf("\nGlobal ID = %d\n", global_id);
printf("Number of semesters = %d\n", m);
printf("Cumulative credits = %d\n", cumulative_TC);

```

```
printf("CGPA = %.2f\n", CGPA);  
  
return 0;  
}
```

## 5 Compilation

```
gcc -std=c11 -Wall -Wextra -pedantic sgpa_cgpa.c -o sgpa_cgpa
```

Compilation completed successfully with no warnings or errors.

## 6 Execution

```
(kali[kali])-[~/demo1]  
$ gcc -std=c11 -Wall -Wextra -pedantic sgpa_cgpa.c -o sgpa_cgpa  
(kali[kali])-[~/demo1]  
$ ./sgpa_cgpa  
Enter global ID: 590033602  
Enter number of semesters: 2  
  
Enter number of courses in semester 1: 2  
Course 1: code 101, credits 3, grade 0  
Course 2: code 102, credits 4, grade A  
Semester 1 SGPA = 9.43  
  
Enter number of courses in semester 2: 3  
Course 1: code 201, credits 3, grade B  
Course 2: code 202, credits 2, grade C  
Course 3: code 203, credits 4, grade A  
Semester 2 SGPA = 8.22  
  
Global ID = 590033602  
Number of semesters = 2  
Cumulative credits = 16  
CGPA = 8.75
```

Figure 1 Two semester SGPA and CGPA execution

## 7 Calculation verification

Semester 1:  $(10 \times 3 + 9 \times 4) / 7 = 66 / 7 = 9.43$ . Semester 2:  $(8 \times 3 + 7 \times 2 + 9 \times 4) / 9 = 74 / 9 = 8.22$ . CGPA:  $(66 + 74) / (7 + 9) = 140 / 16 = 8.75$ .

## 8 Conclusion

The program correctly validates all required inputs, calculates the SGPA for each semester, and calculates the cumulative credit-weighted CGPA. Every prepared test case produced the expected result.